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Deep learning for anomaly detection in traditional Tibetan timber structures: A VAE-based model with multi-head attention and LSTM encoding and state-based thresholding

Xiaowen Qian [a], Na Yang [a,b,*]

[a] School of Civil Engineering, Beijing Jiaotong University, Beijing 100044, China
[b] Beijing's Key Laboratory of Structural Wind Engineering and Urban Wind Environment, Beijing, Jiaotong University, Beijing 100044, China

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ABSTRACT

Recently, there has been growing interest in leveraging deep learning methods combined with static monitoring data, such as tilt angles and strain measurements, to assess the condition of timber structures in historic buildings. In the field of deep learning, anomaly detection is typically performed by comparing real data with its reconstructed data. The larger the error, the higher the likelihood of structural anomalies. This paper first proposes a strategy for constructing high-quality static monitoring datasets free from anomalies. The strategy involves removing outliers using a hybrid Sigma-Local Outlier Factor algorithm and filling missing values with the NeuralProphet algorithm. To enhance the quality of reconstructed data, this paper proposes an LSTM-MHSA-VAE neural network by integrating the Long Short-Term Memory network and the Multi-Head Self-Attention mechanism within the framework of a Variational Autoencoder. Finally, to improve the accuracy of anomaly detection, this paper proposes a state-based threshold determination method. If the error exceeds this threshold, the structure is considered to exhibit an anomaly at that time step. The method proposed in this paper is applied to anomaly detection in traditional Tibetan timber structures, focusing on the tilt angles and strain parameters of the timber components. The results show that, compared to existing methods, the proposed approach demonstrates higher accuracy in anomaly detection and diagnosis.

1. Introduction

The opening of the Qinghai-Tibet Railway in July 2006 great boosted Tibet's tourism industry. However, the rapid growth and the increasing number of visitors have raised serious concerns about the safety of Tibetan heritage structures. For example, renowned heritage sites like the Potala Palace and Jokhang Temple in Lhasa experience excessive loads from millions of visitors each year, resulting in noticeable deformations in structural components [1]. The continuous increase in visitor numbers intensifies structural stress, accelerates material degradation, and heightens the risk of potential failures. Therefore, developing robust anomaly detection methods for timely assessment and preservation has become essential. The design information of these historical structures does not exist or may even have been lost. Consequently, Structural Health Monitoring (SHM) has become a feasible approach to assessing their conditions for preventive conservation [2,3]. While existing SHM technologies can easily collect operational data on structures and their environments, the challenge lies in accurately interpreting these data to assess the structural state.

Structural anomaly detection methods based on SHM data mainly include statistical methods, machine learning algorithms, and deep learning-based methods. Statistical methods primarily rely on probabilistic approaches to detect anomalies by constructing accurate mathematical models based on monitoring data [4]. These models describe normal structural behavior statistically, enabling the identification of deviations that may suggest potential damage or abnormal conditions. For example, probability distribution models like Bayesian networks [5] and Gaussian mixture models [6] are often used to represent the underlying uncertainty in the system, which can then be used to detect anomalies. For instance, Zhang et al. [5] proposed an online anomaly detection approach for structural health monitoring data using a Bayesian dynamic linear model, combining expectation maximization and Kalman smoothing to identify and reconstruct anomalies efficiently. This method was applied to real bridges for anomaly detection, and its

* Corresponding author at: School of Civil Engineering, Beijing Jiaotong University, Beijing 100044, China.
E-mail address: nyang@bjtu.edu.cn (N. Yang).

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effectiveness was successfully validated. However, the limitations of statistical methods include their reliance on prior assumptions about data distribution, limited generalization ability for complex or evolving data, poor handling of high-dimensional data, lack of robustness against noise and missing data, and sensitivity to threshold selection.

Machine learning algorithms, such as clustering, density estimation, distance-based methods, and classification techniques, are commonly employed for anomaly detection. Alireza Entezami et al. [7,8] proposed two methods for structural anomaly detection: one combining manifold learning-assisted clustering (RGMM-NNG) with non-parametric probabilistic methods for anomaly monitoring and early damage warning, and another using an unsupervised learning framework based on non-parametric clustering (HIC) and density anomaly detection (DAD) to address early damage warning in bridge structures during short-term monitoring. Ramin Ghiasi et al. [9] introduced an unsupervised anomaly detection framework based on one-class support vector machine. By analyzing the time-domain features of onboard vibration data from high-speed trains, it addressed the real-time monitoring of railway track geometry defects, improving detection accuracy by 12 % compared to traditional methods. Hassan Sarmadi et al. [10] employed three different Mahalanobis distance (MD) methods (UMD, MSD, LMSD) for ensemble learning and combined the distance participation factor standard to remove environmental influences for structural anomaly monitoring. Li et al. [11] achieved real-time monitoring and condition assessment for high-speed railway bridges using multi-band FIR filtering and clustering. While these methods have been widely applied to structural anomaly monitoring with varying degrees of success, the rapid development of SHM systems has resulted in a substantial increase in both data volume and dimensionality. Lei et al. [12] proposed a two-stage anomaly detection framework combining bayesian neural network and MD. This method was applied to real bridges for anomaly detection, and its effectiveness was successfully validated. Consequently, effectively processing such complex and high-dimensional data remains a significant challenge, often limiting the accuracy and reliability of existing methods.

Deep learning-based structural anomaly detection methods can be categorized into vision-based methods and data-driven methods. Vision-based methods primarily rely on images, offering intuitive visual insights. For example，Bao et al. [13] transformed health monitoring data into image and then applied deep neural networks for anomaly classification. This approach was implemented on real bridges, demonstrating high accuracy. However, they face significant challenges related to dataset construction and computational demands during model training. In data-driven approaches, researchers commonly utilize Autoencoders (AE) [14], Variational Autoencoders(VAE) [15,16], Generative Adversarial Network (GAN) [17], Long Short-Term Memory (LSTM) network [18] and others for data reconstruction or prediction. These models are typically trained using monitoring data representing normal patterns. As a result, the reconstructed or predicted data corresponds to normal conditions. Anomaly detection is performed by comparing the reconstructed or predicted data with the actual monitoring data; if the discrepancy between them exceeds a predefined threshold, the structure is considered abnormal. Fabrizio Falchi et al. [19] used the Temporal Fusion Transformer (TFT) to combine environmental parameters and vibration data for structural health monitoring of the Guinigi Tower in Italy, effectively detecting abnormal dynamic responses caused by earthquakes and simulated damage. Given that monitoring data is often time series in nature, with strong temporal dependencies, some researchers have incorporated Recurrent Neural Networks (RNN) [20] or LSTM networks into AE and VAE models for structural anomaly detection. Networks such as LSTM-SDAE [21], MA-LSTM-VAE [22], VAE-LSTM [23], and VAE-TALSTM [24] have been developed for anomaly detection, achieving promising results in practical engineering applications.

Compared to statistical methods and machine learning algorithms, deep learning-based approaches have shown strong capabilities in structural anomaly detection, particularly for processing high-dimensional SHM data. However, many studies have primarily focused on anomaly detection using acceleration data, which is often assumed to follow a Gaussian distribution in certain conditions. In contrast, the long-term static monitoring data of timber structures (e.g., tilt angles and strain) often exhibit multiple periodicities (daily and annual cycles), mainly influenced by temperature and humidity variations, while creep behavior may contribute to gradual long-term trends. Existing neural networks struggle to fully capture global temporal features, long-term dependencies, and complex data distribution patterns, which can affect anomaly detection performance, particularly for reconstruction-based approaches.

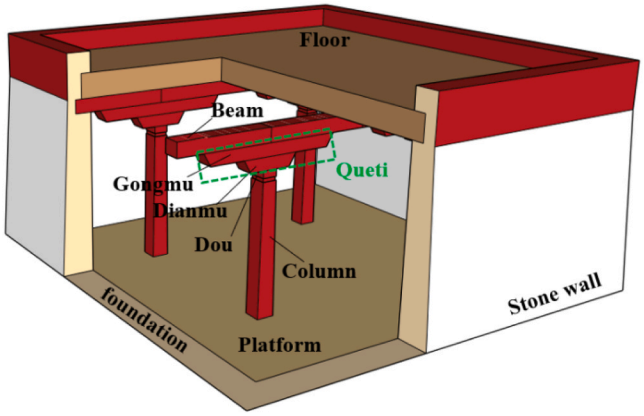


Fig. 1. Traditional Tibetan architecture.

Furthermore, limitations in threshold determination methods can affect anomaly detection precision. These methods include empirical approaches that rely on expert knowledge, statistical methods that use data properties like mean and standard deviation [3], adaptive methods that dynamically adjust thresholds but are sensitive to noise [25], and machine learning-based methods [10]. While machine learning-based methods can enhance detection performance, their accuracy is limited by the difficulty in aligning model architectures with data distribution and extracting high-quality feature representations, particularly when dealing with complex, high-dimensional data. To address these challenges, this paper proposes an LSTM-MHSA-VAE neural network that integrates LSTM and Multi-Head Self-Attention (MHSA) mechanisms within the VAE framework to overcome these shortcomings. Additionally, a state-based threshold determination method is introduced to further enhance the accuracy of anomaly detection.

This paper mainly focuses on structural anomaly detection of Traditional Tibetan Timber Structures (TTTSs), and abnormal data caused by sensor faults has been filtered out in advance. The primary contributions of this paper are as follows: 1) A hybrid Sigma-LOF algorithm is introduced to eliminate outliers in historical data, and the NeuralProphet (NP) algorithm [26] is employed to address missing values, enabling the construction of a high-quality dataset. 2) An LSTM-MHSA-VAE neural network is proposed to improve data reconstruction accuracy. 3) A state-based threshold determination method, leveraging the characteristics of data distribution, is proposed to further enhance the accuracy of structural anomaly detection.

2. The SHM system and characteristics of monitoring data

2.1. The traditional Tibetan timber structures

Traditional Tibetan architecture is characterized by a load-bearing stone wall system combined with a timber frame structure [27], as shown in Fig. 1. TTTSs consist of the confined space defined by the stone walls, which bear the majority of the weight from the floors above. The timber column, Dou, Dianmu, Gongmu, beam, and other components

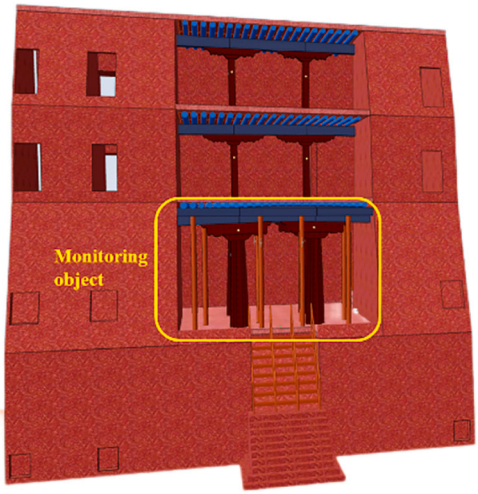


Fig. 2. Monitored structure.

constitute the supporting column assembly. More details on the configuration of the construction can be found in [28].

The structure studied in this paper is a timber temple building within a traditional Tibetan architectural complex located in Lhasa, the capital of the Tibet Autonomous Region of China. Schematic diagrams of the building are shown in Fig. 2. This structure is selected due to its representative structural form, component configuration, connections and materials, which characterize TTTSs.

2.2. Structural health monitoring system

The SHM system of the TTTS comprises both hardware and software subsystems, as illustrated in Fig. 3. The hardware subsystem includes sensors that collect structural data, which is then processed and converted into ASCII format for storage. The software subsystem manages data backup, conversion, and storage in a database, accessible through a web interface. Additionally, it includes a safety alarm system to alert users to anomalies, with real-time monitoring provided via graphical interfaces displaying the structure's condition.

Fig. 4 shows the sensor type and their location relative to a timber frame. The structure consists of four main columns (C1, C2, C3, and C4), six main beams (B1, B2, B3, B4, B5, and B6), and eight reinforcing timber columns (JG1, JG2, JG3, JG4, JG5, JG6, JG7, and JG8). The static variables monitored by the SHM system include the strain (S), tilt angle (TA), relative temperature (RT), ambient temperature (AT) and ambient humidity (AH). Since the system was installed in different stages, the monitored data are labelled as from Stage A and Stage B with 11 and 3 years of data collected, respectively. The fiber grating sensor

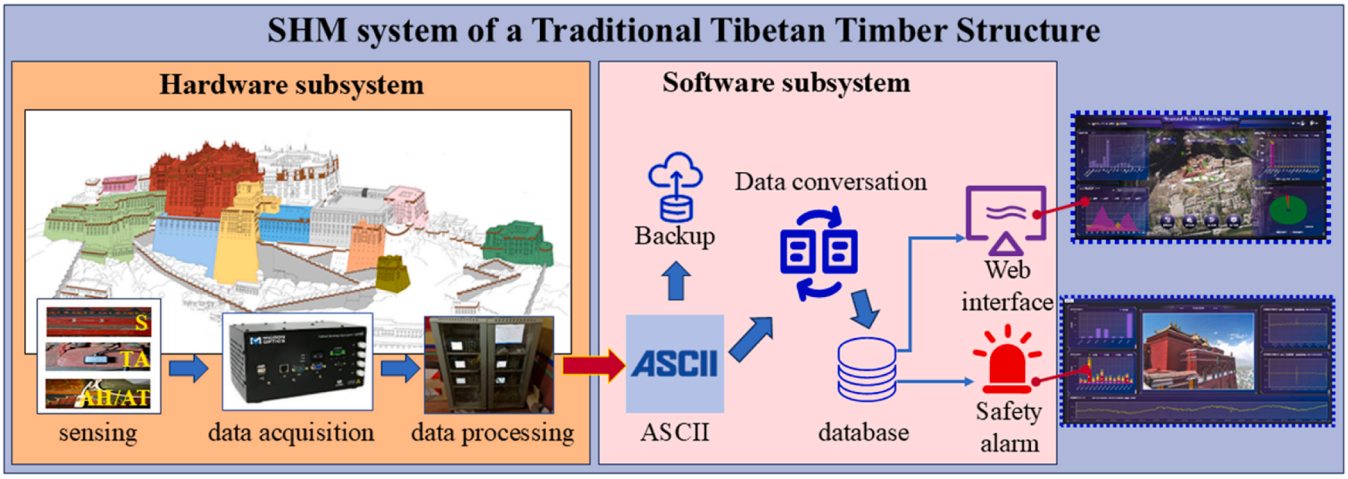
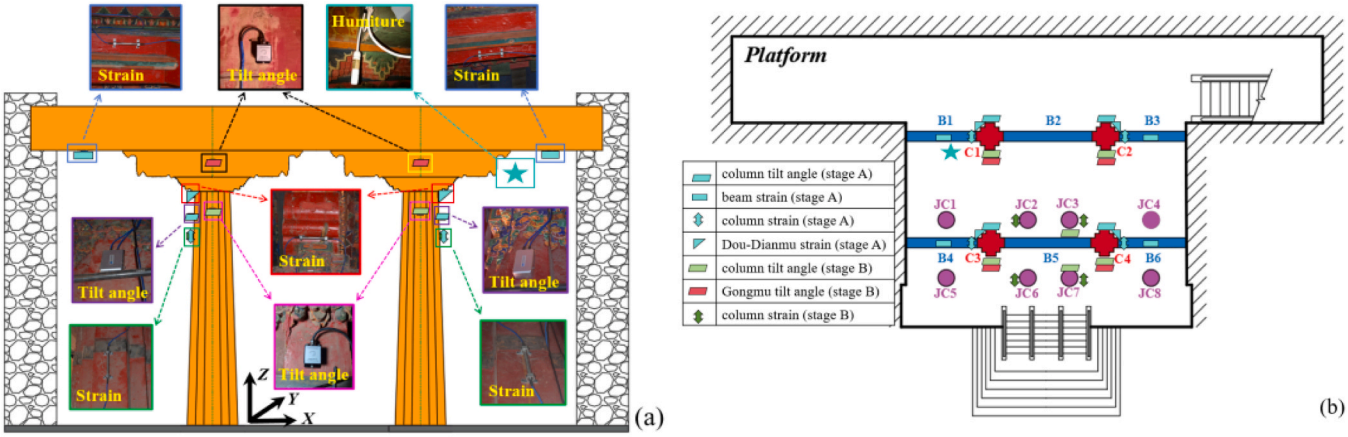


Fig. 3. Topology diagram of a SHM system for TTTSs.



(a) out-of-plane view. (b) plan view.

Fig. 4. Sensor configuration.

Table 1
Parameters of TTTS SHM system.

IDX	Com.	Var.	Unit	Data range	IDX	Com.	Var.	Unit	Data range
1	C1	**S**	**με**	**[−1500,1500]**	**31**	B6	**S**	**με**	**[−1500,1500]**
2		**RT**	**℃**	**[−40,100]**	32		RT	℃	[−40,100]
3		**TA-X**	**°**	**[−5,5]**	33	C1	TA-X	°	[−5,5]
4		RT	℃	[−40,100]	34		TA-Y	°	[−5,5]
5	DC1	**S**	**με**	**[−1500,1500]**	35	G1	TA-X	°	[−5,5]
6		RT	℃	[−40,100]	36		TA-Y	°	[−5,5]
7	C2	**S**	**με**	**[−1500,1500]**	37	C2	TA-X	°	[−5,5]
8		RT	℃	[−40,100]	38		TA-Y	°	[−5,5]
9		**TA-X**	**°**	**[−5,5]**	39	G2	TA-X	°	[−5,5]
10		RT	℃	[−40,100]	40		TA-Y	°	[−5,5]
11	DC2	**S**	**με**	**[−1500,1500]**	41	C3	TA-X	°	[−5,5]
12		RT	℃	[−40,100]	42		TA-Y	°	[−5,5]
13	C3	**S**	**με**	**[−1500,1500]**	43	G3	TA-X	°	[−5,5]
14		RT	℃	[−40,100]	44		TA-Y	°	[−5,5]
15		**TA-X**	**°**	**[−5,5]**	45	C4	TA-X	°	[−5,5]
16		RT	℃	[−40,100]	46		TA-Y	°	[−5,5]
17	DC3	**S**	**με**	**[−1500,1500]**	47	G4	TA-X	°	[−5,5]
18		RT	℃	[−40,100]	48		TA-Y	°	[−5,5]
19	C4	**S**	**με**	**[−1500,1500]**	49	C6	TA-X	°	[−5,5]
20		RT	℃	[−40,100]	50		TA-Y	°	[−5,5]
21		**TA-X**	**°**	**[−5,5]**	51	C8	TA-X	°	[−5,5]
22		RT	℃	[−40,100]	52		TA-Y	°	[−5,5]
23	DC4	**S**	**με**	**[−1500,1500]**	53	C5	S	με	[−1500,1500]
24		RT	℃	[−40,100]	54	C6	S	με	[−1500,1500]
25	B1	**S**	**με**	**[−1500,1500]**	55	C7	S	με	[−1500,1500]
26		RT	℃	[−40,100]	56	C8	S	με	[−1500,1500]
27	B3	**S**	**με**	**[−1500,1500]**	57	C2	AH	℃	[−40,100]
28		RT	℃	[−40,100]	58	C2	AT	%	[0,100]
29	B4	**S**	**με**	**[−1500,1500]**	-	-	-	-	-
30		RT	℃	[−40100]	-	-	-	-	-

Note: Bolded font is the parameter chosen for model training.

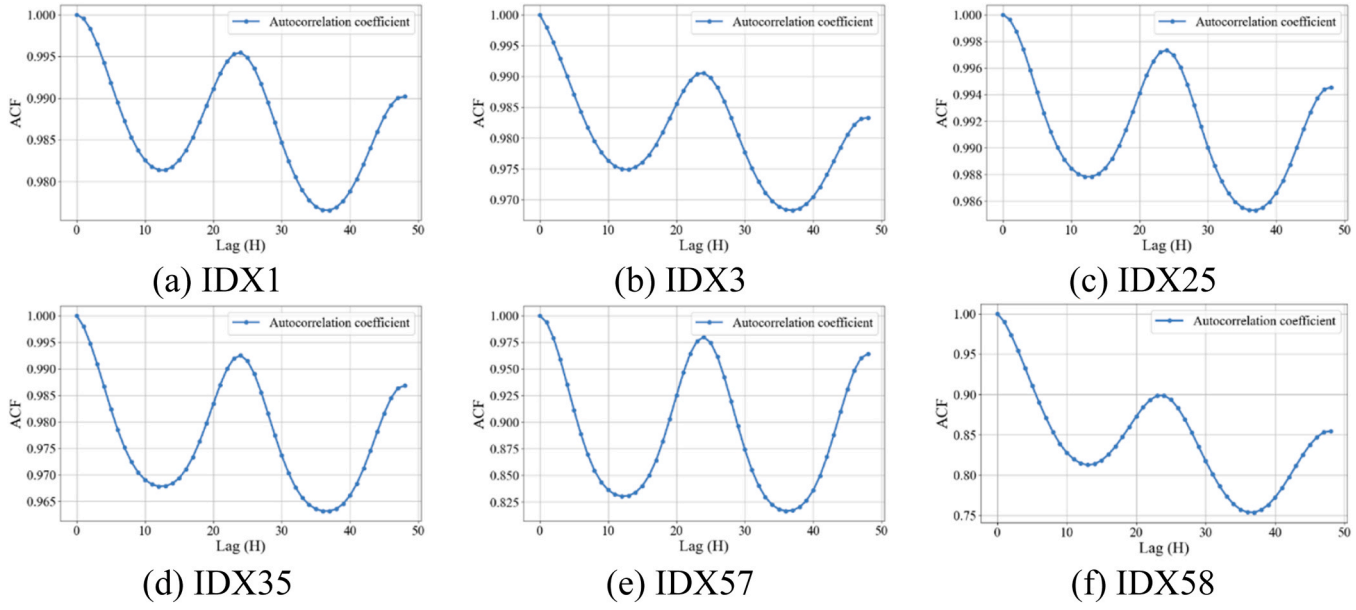


Fig. 5. The variation curves of autocorrelation coefficients for different parameters.

installed in Stage A adopts a wired acquisition mode, with strain sensors and tilt angle sensors. Each sensor collects one sample of two types of data (S-RT or TA-RT) at 5 minutes interval. Unconventional instruments such as MEMS dipmeter, vibrating wire strain gauge, and hygro-thermograph have been installed in Stage B. Additionally, a wireless data collection method was installed in Stage B to collect one sample from all sensors at 60 minutes interval. Table 1 provides a summary of the monitored information to facilitate subsequent data analysis in this report. Labels IDX1-IDX32 and IDX33-IDX58 denote the monitored parameters in Stage A and Stage B, respectively.

2.3. Characterization of TTTSs monitoring data

2.3.1. Time-dependent properties in both long and short terms

In this study, the time-dependent characteristics of SHM data are analyzed using both the autocorrelation coefficient function (ACF) and the Hurst exponent[29]. The ACF is used to assess short-term dependencies by examining the correlation between data points at different time lags. A significant autocorrelation at small lags indicates that the current measurement is closely related to the previous one, revealing the presence of short-term memory or persistence in the data.

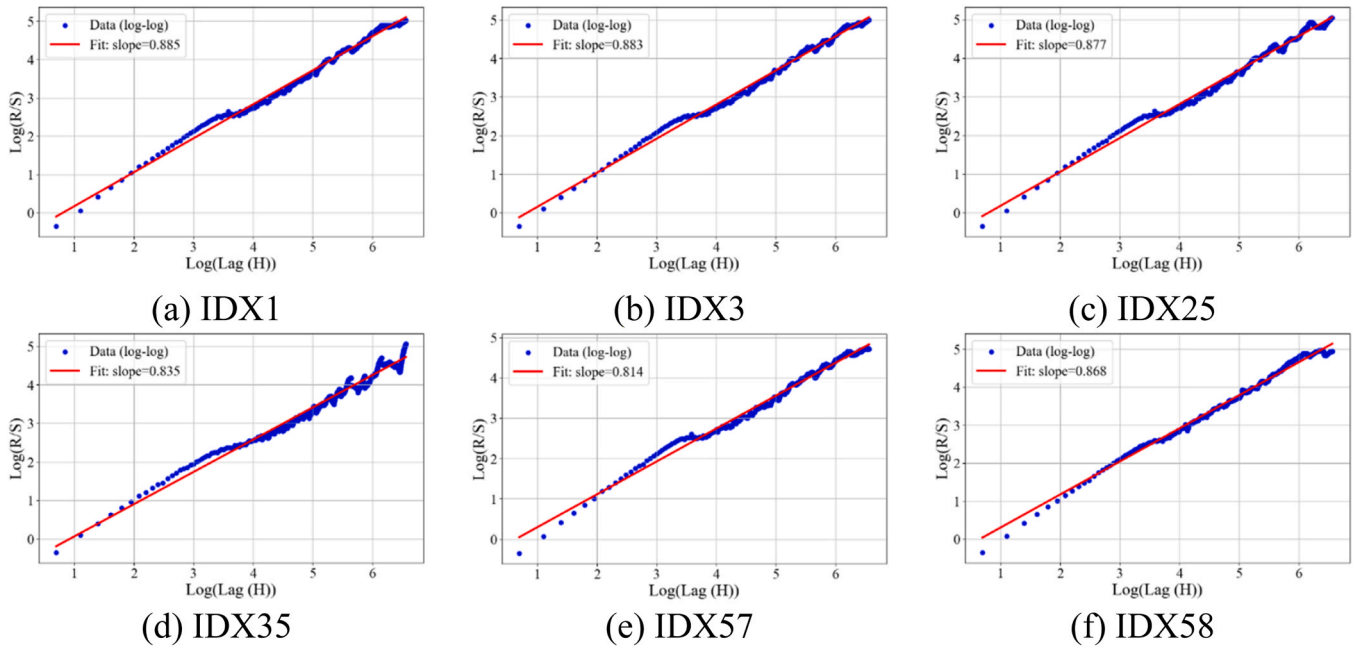


Fig. 6. The fitted curves of Hurst exponents for different parameters.

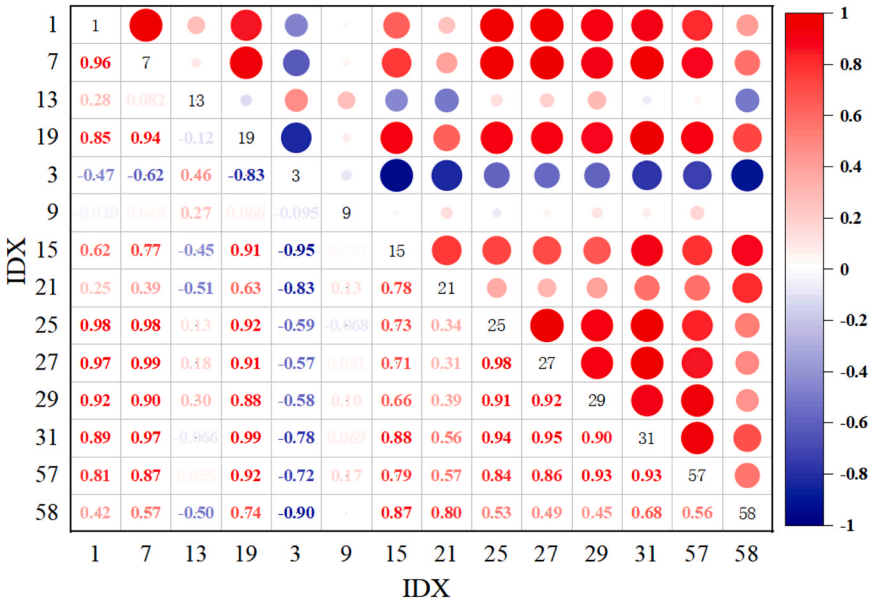


Fig. 7. Correlation coefficient plot.

In contrast, the Hurst exponent is used to assess long-term dependencies by quantifying the degree of long-term memory in the time series. A Hurst exponent value greater than 0.5 suggests long-term persistence, meaning the series tends to follow its past behavior, while a value less than 0.5 indicates anti-persistence, where the series reverts after a deviation.

An analysis was conducted on the monitoring data collected between January 1, 2022, and December 31, 2023, with a sampling frequency of one measurement per hour. Data from six monitoring points—IDX1, IDX3, IDX25, IDX35, IDX57, and IDX58—were selected, corresponding to six parameters: column strain, column tilt angle, beam strain, Gongmu tilt angle, ambient temperature, and ambient humidity, respectively. Fig. 5 illustrates the autocorrelation coefficients for each parameter over the time interval [0,48] hours with varying lags. It can be observed that, within a time interval of 2 days (48 hours), the autocorrelation coefficients for all parameters remain above 0.7, indicating significant short-term dependence in the monitoring data. Fig. 6 presents the fitted curves of the Hurst exponent for each parameter. The calculated Hurst exponents for IDX1, IDX3, IDX25, IDX35, IDX57, and IDX58 are 0.885, 0.883, 0.877, 0.835, 0.817, and 0.868, respectively. Since all Hurst exponents exceed 0.5, the monitoring data from TTTSs demonstrate a notable long-term dependence. In conclusion, the TTTSs monitoring data exhibit both short-term dependence (strong correlation within a 2-day interval) and long-term dependence (Hurst exponent > 0.5). These findings provide valuable insights into the characteristics of the monitored data and form a basis for further investigations.

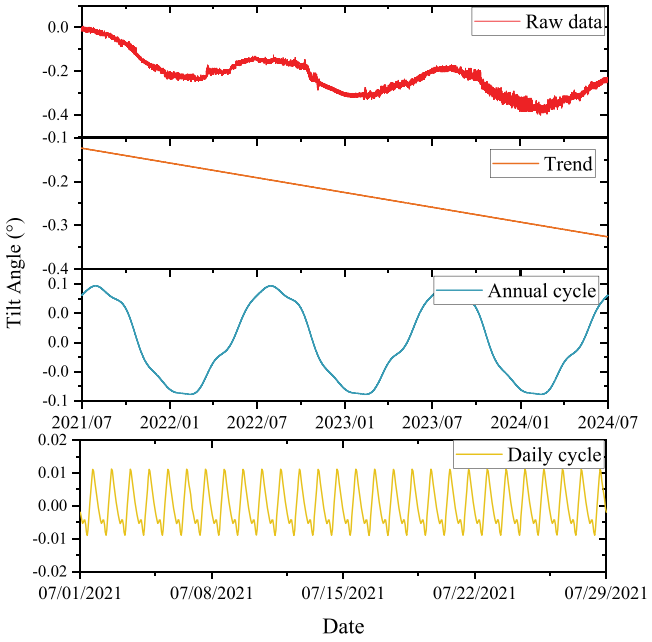


Fig. 8. Schematic diagram of components in IDX33 decoupled by NP model.

2.3.2. Correlation analysis

Considering the large number of parameters in Table 1, this section selects representative data for correlation analysis. The selected data includes four timber columns strain data (IDX1, IDX7, IDX13, IDX19), four timber columns tilt angle data (IDX3, IDX9, IDX15, IDX21), four beams strain data (IDX25, IDX27, IDX29, IDX31), and environmental temperature and humidity data (IDX57, IDX58). Fig. 7 shows the correlation coefficient matrix among these variables. Key observations from the figure are as follows:

1) Strong correlation between strain and temperature: Most strain data exhibit correlation coefficients above 0.8 with temperature, indicating a significant influence of temperature on strain.

2) Weak correlation between strain and humidity: The correlation coefficients between strain and humidity are generally below 0.7, suggesting a limited impact of humidity on strain.

3) Poor correlation between strain and tilt angle: The correlation between strain and tilt angle data is generally low, indicating no significant relationship between them.

2.3.3. Data distribution characteristics

In this section, NP algorithm is employed to decompose monitoring data into trend, seasonality, and residual components. This decomposition reveals that if the data exhibits clear periodicity, it suggests that the distribution characteristics are not static but rather evolve over time. Specifically, the data's mean and variance may fluctuate across different time intervals, reflecting dynamic distribution changes. To further explore these distributional properties, Gaussian Mixture Model (GMM) [30] is applied for data fitting, capturing the multimodal nature of the data. GMM allows for the quantification of the distribution's temporal dynamics by estimating the parameters of the Gaussian components, such as mean, variance, and weight, over time. Combining the results from NP decomposition and GMM fitting provides a comprehensive understanding of the dynamic and evolving distribution of the data, offering valuable insights into its temporal dependencies and underlying patterns.

Fig. 8 illustrates the decoupling results of IDX33 over a three-year period (July 2021–July 2024) using the NP algorithm. The subplots, from top to bottom, represent the original data, trend component, annual cycle component, and daily cycle component. The trend component reveals that the tilt angle of the timber column changed by approximately 0.3° over this period. The annual cycle component indicates that seasonal variations cause the tilt angle to fluctuate within the range of [-0.1°, 0.1°], while the daily cycle component demonstrates that daily temperature variations lead to fluctuations within the range of [-0.01°, 0.01°]. These findings suggest that the data distribution is influenced by both annual and daily cycles.

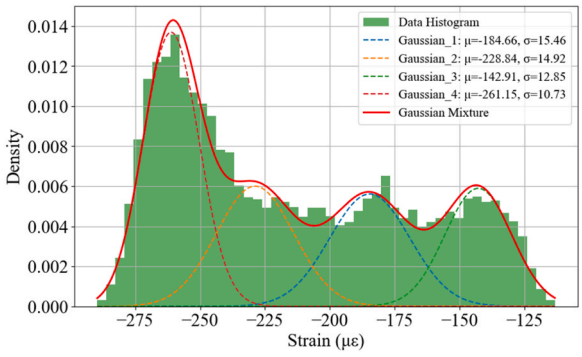


Fig. 9. GMM fitting of IDX1.

Fig. 9 depicts the GMM fitting of IDX1 data, which successfully decomposes the data into four individual Gaussian distributions. However, traditional GMMs face limitations in handling complex distribution characteristics, as they struggle to effectively capture the latent structures and high-dimensional nonlinear relationships in the data.

To address these challenges, this study introduces the VAE [31] model. With its powerful nonlinear modeling capabilities and ability to learn latent representations, the VAE is employed to further explore the latent distribution characteristics of TTTS monitoring data. The VAE excels at learning the latent distribution of complex data in an unsupervised manner, particularly in capturing multimodal features akin to Gaussian mixtures. This provides a novel perspective and methodological support for analyzing the data's distributional characteristics.

3. The flow of anomaly detection

3.1. Problem statement

This section introduces the basic formulation of the problem studied. Lowercase italicized letters, e.g., x, denote scalars and uppercase italicized letters, e.g., X, denote vectors. Bold capital letters, e.g., $\mathbf{X}$, denote matrices. The monitoring data collected from the TTTS is represented by a set of time series $\mathbf{X} = \langle X^1, X^2, X^3, \cdots, X^K \rangle = \langle X_1, X_2, X_3, \cdots, X_L \rangle^T \in \mathbb{R}^{L \times K}$, where superscript K denotes the number of parameters, and superscript L denotes the total number of time steps. Specifically, $X^i = (x_1^i, x_2^i, x_3^i, \cdots, x_L^i)$ represents the data sequence of the ith parameter.$X_t = (x_t^1, x_t^2, x_t^3, \cdots, x_t^K)$ is a K-dimensional vector describing the state of the TTTS at time step t.

The problem studied is defined as follows: Given a set of SHM data$\mathbf{X}$, the objective is to calculate the abnormal score s of the vector X_t. The abnormal score is obtained by solving the MD [32] between the reconstructed and original data. The higher the abnormal score, the more likely that vector X_t is anomalous. A threshold η is defined that if the abnormal score $s > \eta$, vector X_t is considered to be anomalous. The focus of this paper is to assess the structural health of TTTSs by monitoring tilt angles and strain data to identify potential deviations in structural behavior.

3.2. Framework of the proposed method

A structural anomaly detection framework is proposed, as shown in Fig. 10. It includes the core stages of data preprocessing, neural network training and the determination of state-based thresholds. They are described in detail as follows.

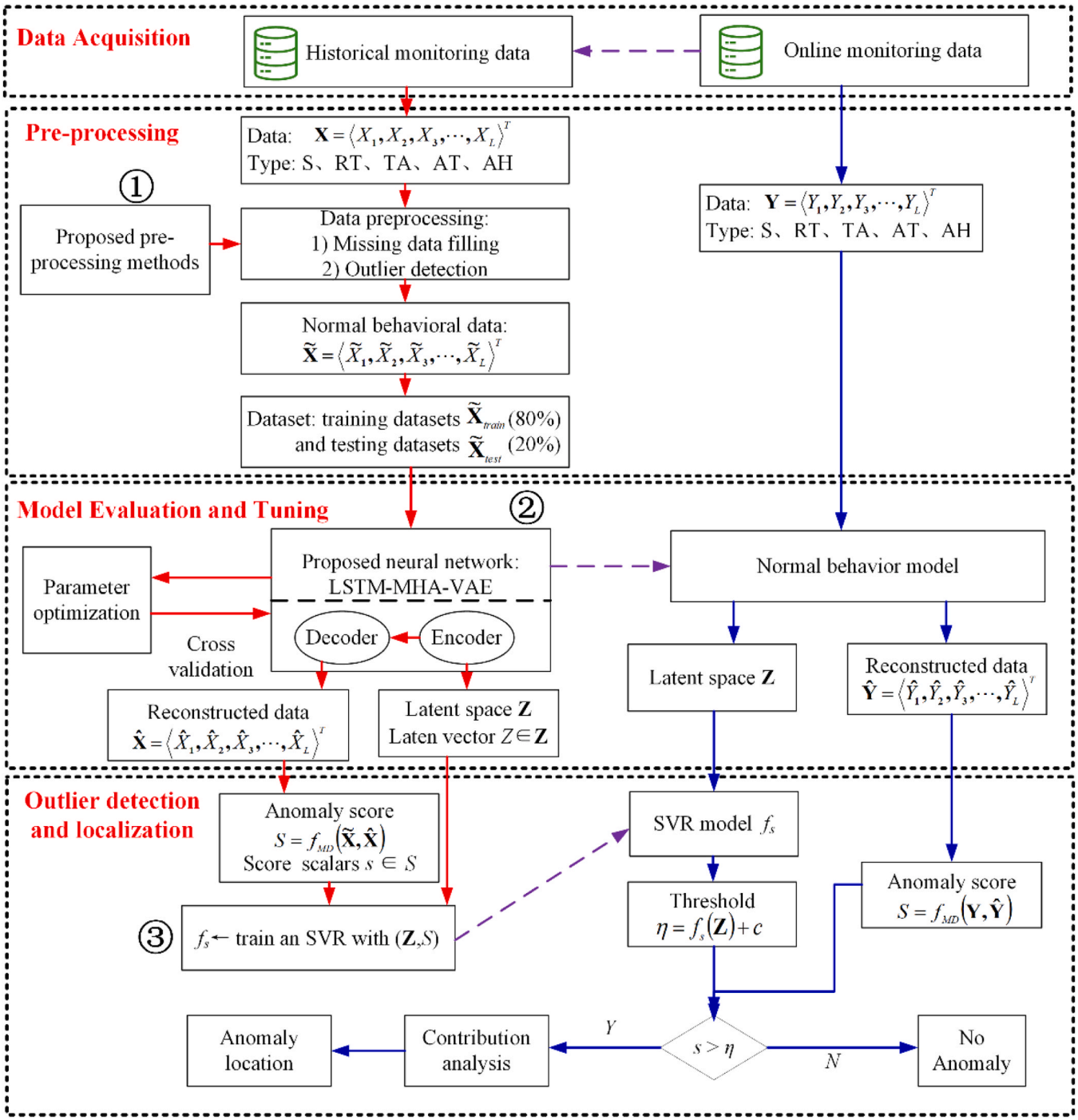


Fig. 10. Structural anomaly detection framework.

(1) **Data Collection**. The monitoring data collected from the TTTS is referred to as historical data, denoted as $\mathbf{X}=\langle X^1,X^2,X^3,\cdots,X^K\rangle=\langle X_1,X_2,X_3,\cdots,X_L\rangle^T$, while the data that are currently being collected or will be collected in future are referred to as online data, denoted as $\mathbf{Y}=\langle Y^1,Y^2,Y^3,\cdots,Y^K\rangle=\langle Y_1,Y_2,Y_3,\cdots,Y_L\rangle^T$.

(2) **Data Preprocessing**. During the data collection process, outliers and missing values may occur. It is then preprocessed to obtain data $\widetilde{\mathbf{X}}$ with a normal pattern for training the proposed neural network. Detailed procedure on the data preprocessing is described in Section 4.1.

(3) **Train the LSTM-MHSA-VAE neural network**. LSTM-MHSA-VAE is a novel neural network proposed in the paper to improve the quality of reconstructed data $\hat{\mathbf{X}}$. Its novelty lies in the integration of LSTM layers and MHSA mechanisms within the VAE framework. The LSTM-MHSA-VAE neural network is described in detail in Section 4.2.

(4) **The anomaly score**. The MD between the training data, $\widetilde{\mathbf{X}}_{train}$, and the reconstructed data, $\hat{\mathbf{X}}$, is taken as the anomaly score, S, where each individual anomaly score, $s\in S$, represents the score for a specific time step. The calculation for the MD is discussed in Section 4.3.

(5) **State-based thresholds**. Support Vector Regression (SVR) [33] is adopted to map the latent vector $\mathbf{Z}$ to the anomaly score s to obtain an anomaly score estimator f_s. Define the state-based threshold as $\eta=f_s(\mathbf{Z})+c$, with c as a constant. The threshold η can be subjectively adjusted by varying c for detecting outliners in data with different measurements.

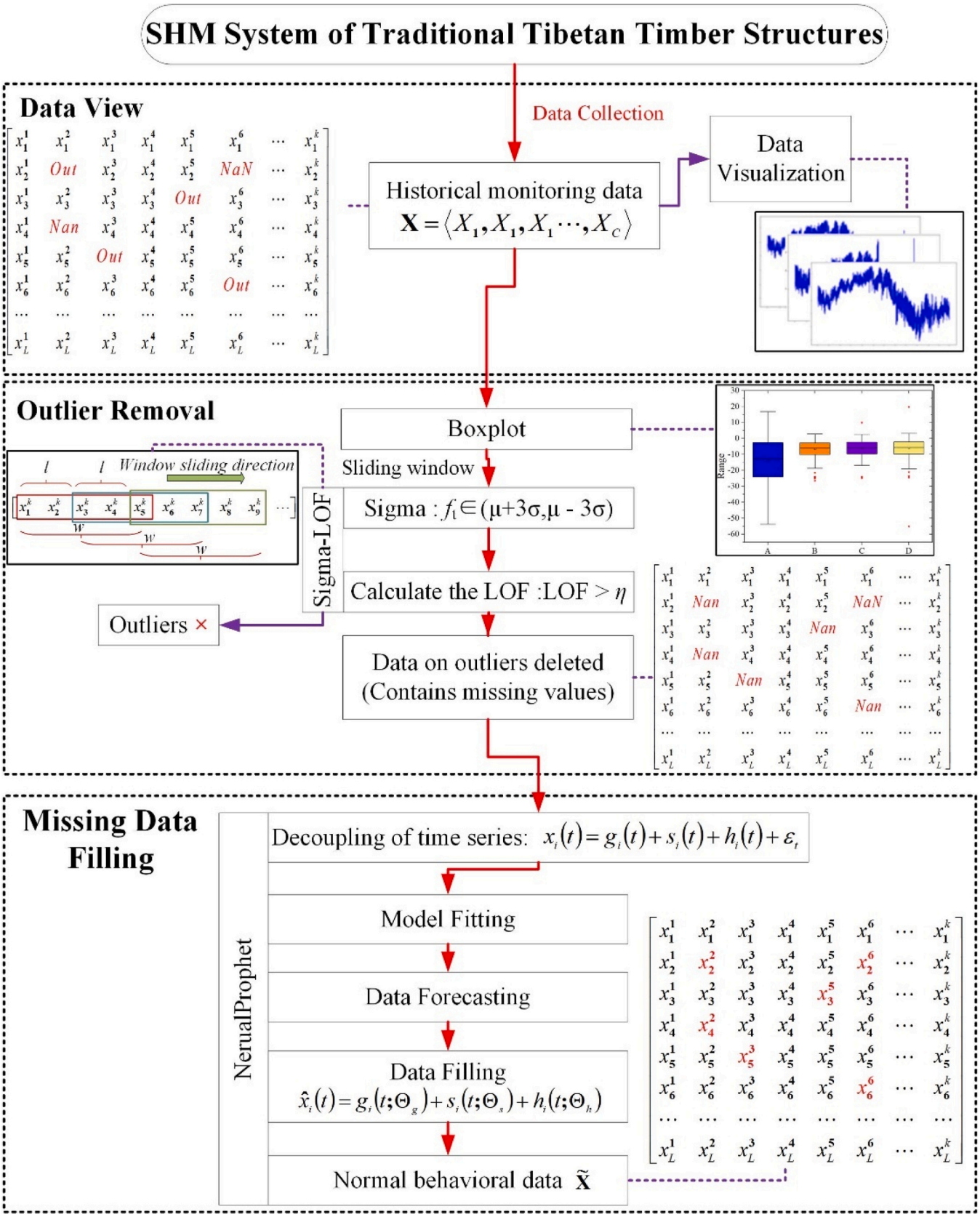


Fig. 11. Framework on preprocessing the monitoring data.

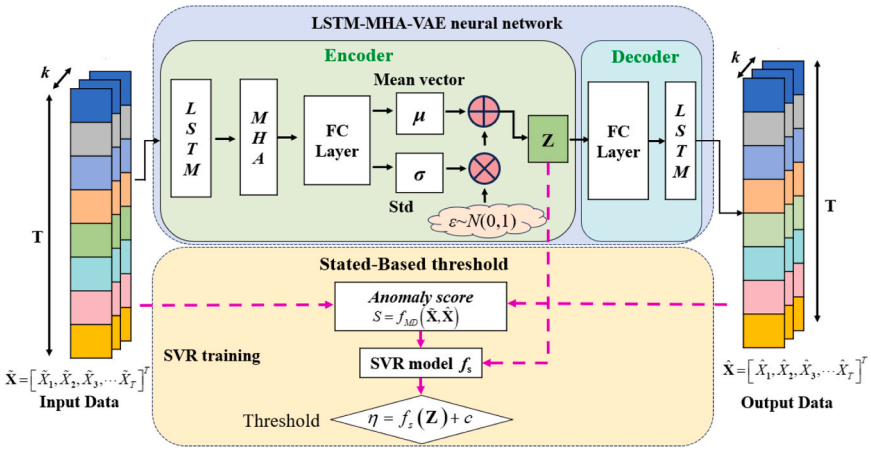


Fig. 12. Proposed LSTM-MHSA-VAE neural network and stated-based threshold.

(6) **Outlier localization**. The above procedure identifies time steps with anomalous data but does not determine which parameter is responsible. This issue is addressed by conducting contribution analysis, as discussed in Section 4.4.

3.3. Criteria for evaluation

3.3.1. Criteria for evaluating the performance of data reconstruction

Two statistical metrics are employed to quantify the discrepancy between the observed and reconstructed data: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). In general, smaller values of these metrics indicate better reconstruction performance [34]. Their mathematical definitions are as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{2}$$

where n denotes the number of data samples, $\hat{y}_i$ denotes the ith reconstructed data, and $y_i$represents the ith observed data.

3.3.2. Criteria for anomaly detection

Several standard metrics have been commonly used for performance evaluation of anomaly detection in machine learning [35], and they are:

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\mathrm{F1Score} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{5}$$

where *TP* - True Positive which is the actual abnormal data identified as abnormal (correct). *FN* - False Negative which is the actual abnormal data identified as normal (incorrect). *FP* - False Positive which is the normal data identified as abnormal (incorrect).

The precision, recall, and F1 score metrics have values between 0 and 1. A value of 1 indicates the best model performance, while a value of 0 indicates the worst. The closer they are to 1, the better the performance.

4. The proposed MHSA-LSTM-VAE neural network

4.1. Preprocessing SHM data

The data obtained from SHM systems of actual engineering structures often face issues such as outliers and missing values, which can lead to the deterioration of time-series features. Therefore, it is necessary to process the data to improve the quality of the dataset and enhance the accuracy and reliability of the training model.

The data preprocessing flowchart is shown in Fig. 11, and the detailed description is as follows:

(1) **Visualization of monitoring data.** The monitoring data **X** obtained from the SHM system is taken as input. Each dimension X^i is plotted to visually identify outliers and missing data.

(2) **Removal of outliers using box plot.** Outliers identified above are removed using box plots [21]. This is an efficient method to eliminate outliers that are significantly distant from the normal data.

(3) **Further screening of outliers using the sigma-LOF algorithm with a sliding window technique.** First, each parameter's monitoring data is divided into a series of data subsets using a window size w and a sliding step l. Then, outliers in each parameter's data subset are eliminated using the 3-sigma rule [36]. Finally, the corrected dataset with outliers removed is rearranged into multiple 2D data subsets, and the LOF algorithm is applied in the spatial domain for further outlier removal. These 2D data subsets represent the data collected by individual sensors. For example, each sensor in Stage A collects response and relative temperature data, while each sensor in Stage B collects the tilt angles of structural components in the X and Y directions, as shown in Table 1.

(4) **Imputation of missing data.** Missing data is imputed using the NP algorithm [37], with the corresponding code provided in [38].

4.2. Construction of LSTM- MHSA -VAE neural network

Section 2.3 analyzes the long-term and short-term dependency characteristics of TTTSs monitoring data, noting that its distribution changes over time, and the data is suitable for GMM modeling. Based on these characteristics, this paper introduces the LSTM-MHSA-VAE neural network, which combines LSTM for learning short-term dependencies, MHSA for capturing long-term dependencies [39], and VAE for modeling data distribution to enhance reconstruction capability (see Fig. 12). The network consists of an encoder and a decoder, where the encoder sequentially integrates LSTM, MHSA, and VAE modules, while the decoder utilizes LSTM to reconstruct the original data from the latent representation. Detailed functionalities of each module, including their inputs, outputs, and training process, are discussed in the following sections.

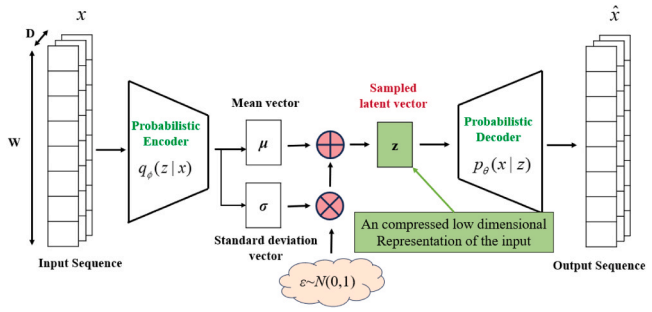


Fig. 13. VAE Model Architecture.

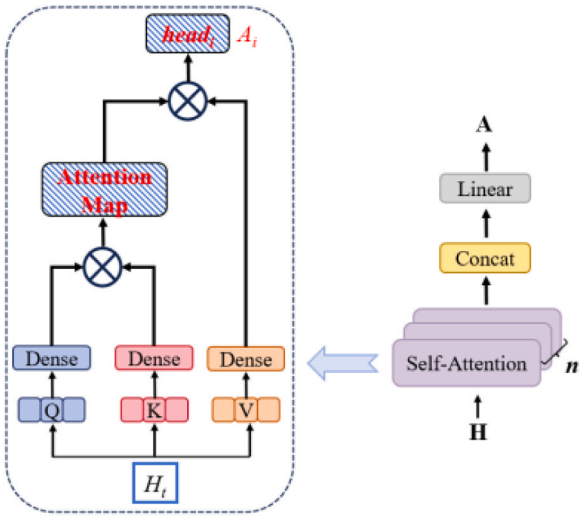


Fig. 14. MHSA mechanism.

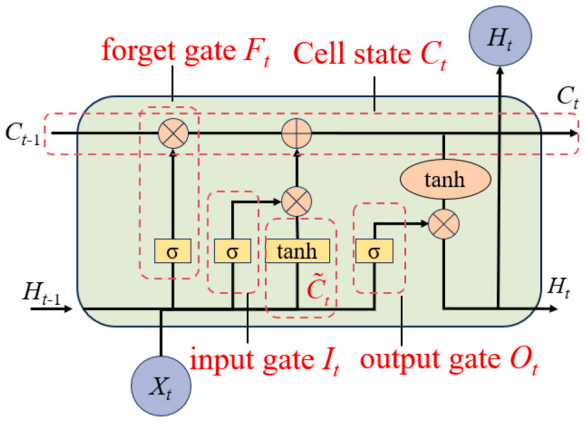


Fig. 15. Structure of memory cell in LSTM.

4.2.1. VAE model

VAE is a generative model designed to learn the underlying distribution of complex data in an unsupervised manner, as shown in Fig. 13. VAE consists of an encoder and a decoder. The encoder maps the input data to a latent space, while the decoder reconstructs the data from this latent representation. The key advantage of VAE is its ability to generate a smooth and continuous latent space, making it suitable for data reconstruction tasks. By imposing a probabilistic framework, VAE can efficiently model the distribution of data, making it particularly useful for learning the complex and multimodal distribution of TTTSs monitoring data. The VAE is trained to minimize the reconstruction loss, along with a regularization term (KL-divergence), ensuring that the learned latent space is both informative and regularized.

4.2.2. Multi-head self-attention mechanism

The MHSA mechanism is a powerful method used to capture dependencies in sequential data, especially in cases where modeling relationships between distant time steps is required, and its architecture is shown in Fig. 14. Unlike RNNs, MHSA does not process data in a sequential manner, allowing it to capture dependencies across all positions in the sequence simultaneously. This is particularly beneficial for learning long-term dependencies that may be difficult to capture with traditional sequence models.

In the MHSA mechanism, each input is assigned attention weights that indicate its importance relative to other elements in the sequence. The multi-head mechanism involves multiple attention operations in parallel, each focusing on different aspects of the data. The outputs of these parallel operations are then concatenated and projected, allowing the model to capture a wide range of dependencies. This makes MHSA highly effective in processing time series data with complex relationships across long intervals, such as TTTSs monitoring data. By incorporating MHSA, our model can efficiently capture the long-term dependencies that play a critical role in understanding the time-varying behavior of the data.

4.2.3. LSTM Neural Network

The LSTM neural network is a variant of RNNs designed to address the vanishing gradient problem in traditional RNNs, as illustrated in Fig. 15. LSTMs are particularly well-suited for capturing both short-term and moderately long-term dependencies in time series data. They use memory cells regulated by gating mechanisms to control the flow of information, enabling the network to model sequential patterns over time. However, while LSTMs can capture long-term dependencies to some extent, they may struggle with very long-range dependencies due to their sequential nature, whereas models like MHSA can directly model global dependencies through parallel attention mechanisms.

As shown in Fig. 15, the LSTM architecture consists of several key components: input gate (I_t), forget gate (F_t), output gate (O_t), and cell state (C_t). The network processes the input X_t and previous hidden state H_{t-1} to update the cell state and generate the current hidden state H_t. The operations within the LSTM cell, including sigmoid (σ) and tanh activations, effectively regulate the flow of information through gating mechanisms.

4.2.4. LSTM-MHSA-VAE Model Training for monitoring data

The training process of the LSTM-MHSA-VAE neural network (as shown in Fig. 12) is described below:

(1) **Data set division.** The preprocessed monitoring data $\widetilde{\mathbf{X}} = \langle \widetilde{X}_1, \widetilde{X}_2, \widetilde{X}_3, \cdots, \widetilde{X}_L \rangle^T$ with missing values filled in and outliers removed is normalized, i.e. scaled or expanded to [0,1]. The data is then divided into training dataset $\widetilde{\mathbf{X}}_{\text{train}} = \langle \widetilde{X}_1, \widetilde{X}_2, \widetilde{X}_3, \cdots, \widetilde{X}_C \rangle^T$ and test dataset $\widetilde{\mathbf{X}}_{\text{test}} = \langle \widetilde{X}_{C+1}, \widetilde{X}_{t+2}, \widetilde{X}_{t+3}, \cdots, \widetilde{X}_L \rangle^T$ in the proportion of 4:1.

(2) **Encoder computation**. Input $\widetilde{\mathbf{X}}_{\text{train}}$into the encoder, and process with LSTM, MHSA and VAE in turn to obtain the latent space **Z.**

(a) LSTM Cell Computations. The training dataset $\widetilde{\mathbf{X}}_{\text{train}}$ is input to the LSTM neural network to obtain the hidden state sequence $\mathbf{H} = \langle H_1, H_2, H_3, \cdots, H_T \rangle$ of the LSTM.

(b) MHSA mechanism. The structure of the MHSA mechanism is shown in Fig. 14. Firstly, given the hidden states **H** from the LSTM, each hidden state is mapped to query, key, and value vectors using linear transformations. Assuming there are n heads, the following steps are performed for each head i:

$$Q^i = W_Q^i \mathbf{H} + b_Q^i \tag{6}$$

$$K^i = W_K^i \mathbf{H} + b_K^i \tag{7}$$

$$V^i = W_V^i \mathbf{H} + b_V^i \tag{8}$$

Where: Q^i, K^i, $V^i$are the matrices containing all queries, keys, and values for the ith head. W_Q^i, W_K^i,and $W_V^i$are the weight matrices for queries, keys, and values, respectively. b_Q^i,b_K^i, and b_V^i are the corresponding bias vectors for each transformation.

The attention scores [40] are computed as:

$$\text{head}_i = \text{Attention}\left(Q^i, K^i, V^i\right) = \text{softmax}\left(\frac{Q^i (K^i)^T}{\sqrt{d_k}}\right) V^i \tag{9}$$

Concatenate the heads to get the final output:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \cdots, \text{head}_h) W_O + b_O \tag{10}$$

Where W_O is the output weight matrix, and b_O is the output bias matrix.

The output of the MHSA mechanism is a new sequence $\mathbf{A} = \langle A_1, A_2, A_3, \cdots, A_T \rangle$.

(c) VAE Encoder. The Mean vector μ and Log-Variance vector $\log \sigma^2$ of sequence $\mathbf{A}$ are calculated as:

$$\mu = W_\mu \mathbf{A} + b_\mu \tag{11}$$

$$\log \sigma^2 = W_\mu \mathbf{A} + b_\sigma \tag{12}$$

The latent space $\mathbf{Z}$ is then obtained as:

$$\mathbf{Z} = \mu + \sigma \odot \varepsilon, \varepsilon \sim N(0, I),$$

where

$$\sigma = \exp\left(\frac{\log \sigma^2}{2}\right) \tag{13}$$

(3) **Decoder computation**. The decoder of the proposed neural network utilizes the LSTM neural network. A fully connected layer is used to convert $\mathbf{Z}$ into an initial hidden state sequence $\widetilde{\mathbf{H}} = \text{Linear}(\mathbf{Z})$, which is used as input to LSTM to obtain the reconstructed data $\widehat{\mathbf{X}} = \text{LSTM}(\widetilde{\mathbf{H}})$.

(4) **Loss Function.** The VAE optimizes the Kullback-Leibler (KL) Divergence for a better understanding of the latent representation and the reconstruction loss to have better reconstructed data. The reconstruction loss measures the discrepancy between the input data and the output from decoder for accurate data reconstruction. The reconstruction loss for the input data $\widetilde{\mathbf{X}}$ and the reconstructed data $\widehat{\mathbf{X}}$ is defined as $L_{reconstruction} = \frac{1}{N}\sum_{i=1}^{n} \|\widetilde{\mathbf{X}} - \widehat{\mathbf{X}}\|^2$.

The KL Divergence measures the difference between the latent distribution $q(Z|X)$ of output from the encoder and the prior distribution $p(Z)$. Prior distribution is usually a standard normal distribution. By minimizing the KL Divergence, the encoder is regularized to generate a latent representation that is conducive to high-quality data generation. For a latent representation with mean μ and variance σ, the KL Divergence is given by $L_{KL} = -\frac{1}{2}\sum_{j=1}^{d_{latent}} \left(1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2\right)$.

The total VAE loss function combines the reconstruction loss and the KL divergence as $loss = L_{reconstruction} + \beta L_{KL}$. Where β is a hyperparameter balancing the two loss terms. Optimizing this total loss function would enable an effective learning of the latent representation of the input data by the VAE to achieve high-quality data reconstruction and generation.

(5) Model training. During the training of the LSTM-MHE-VAE neural network, the weights and biases of network are continuously adjusted using backpropagation and gradient descent optimization methods to minimize the loss function.

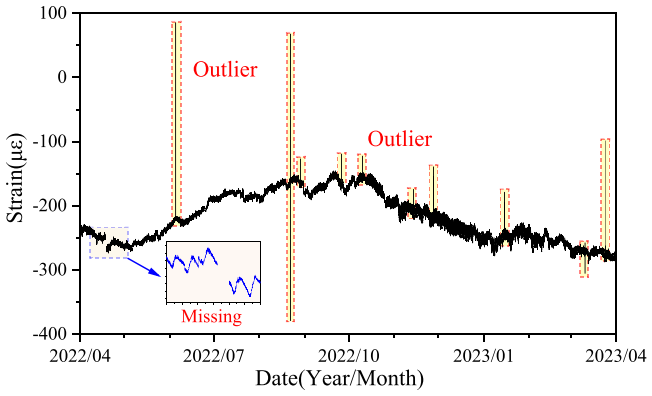


Fig. 16. IDX1 raw data.

4.3. State-based thresholds

The detailed process of state-based threshold determination is shown in Fig. 12. Section 4.2 outlines the training process of the LSTM-MHSA-VAE neural network using preprocessed data $\widetilde{\mathbf{X}}$, which generates the latent space $\mathbf{Z}$ (i.e., state) and the reconstructed data $\widehat{\mathbf{X}}$. It is important to note that the quality of the reconstructed data $\widehat{\mathbf{X}}$ depends on the latent space $\mathbf{Z}$ fed into the decoder [41]. Employing a fixed threshold for anomaly detection may yield inaccurate results for certain states $\mathbf{Z}$. Conversely, adjusting the threshold based on the latent vector $Z \in \mathbf{Z}$ can significantly enhance the accuracy of anomaly detection.

The MD between the input and output data of the LSTM-MHSA-VAE neural network is taken as the anomaly score $\boldsymbol{S}$. The MD between $\widetilde{\mathbf{X}}$ and $\widehat{\mathbf{X}}$ can be calculated as:

$$f_{MD}(\widetilde{\mathbf{X}}, \widehat{\mathbf{X}}) = \sqrt{(\widehat{\mathbf{X}} - \widetilde{\mathbf{X}} - \mu)^T \text{Cov}^{-1} (\widehat{\mathbf{X}} - \widetilde{\mathbf{X}} - \mu)} \tag{14}$$

where μ is the mean vector, and Cov is the covariance matrix of the operating parameters.

Then, an SVR model is constructed to establish the mapping between the latent space Z and the anomaly score S, serving as the anomaly score evaluator f_s. The state-based threshold is defined as: $\eta = f_s(\mathbf{Z}) + c$, where c is a constant that can be adjusted to control the sensitivity of anomaly detection.

The LSTM-MHSA-VAE neural network and SVR model trained from the historical monitoring data can be deployed for anomaly identification of the online monitoring data $\mathbf{Y}$, as shown in Fig. 10. The online monitoring data $\mathbf{Y}$ is inputted into the LSTM-MHSA-VAE neural network to obtain the latent space $\mathbf{Z}$ and the reconstructed data $\widehat{\mathbf{Y}}$. Then, the MD between $\mathbf{Y}$ and $\widehat{\mathbf{Y}}$ is calculated (i.e., anomaly score), and the latent space $\mathbf{Z}$ is input into the trained anomaly score estimator f_s to determine the threshold η. If the anomaly score is greater than the threshold η, the monitoring data at time instant t is considered anomalous, i.e., a structural anomaly.

4.4. Anomaly location

If the monitoring data $Y_t$of the structure produces an outlier at time step t, the anomaly alarm described in Section 4.3 is triggered. It is, however, not known which parameter produces the anomaly, and contribution analysis is needed to locate the parameter that carries the anomaly. More detailed description of anomaly localization can be found in [21].

5. Application to the traditional Tibetan timber structure

This section validates the performances of the proposed LSTM-MHSA-VAE neural network combined with state-based threshold in structural anomaly detection for the TTTS. The dataset for training the LSTM-MHSA-VAE neural network contains a total of 17 parameters which are strain, tilt angle, relative temperature, ambient temperature

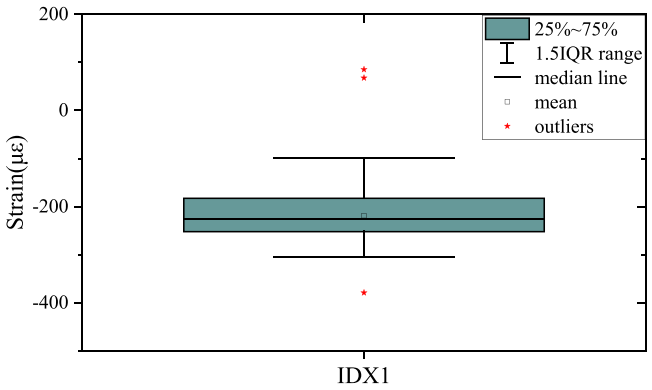


Fig. 17. Box plot.

and humidity, and the 17 parameters are denoted in bold font in Table 1. The time span of this dataset is from April 1, 2022, to March 31, 2023.

5.1. Data preprocessing and dataset construction

The performance of the sigma-LOF algorithm and the NP algorithm for outlier removal and missing data imputation in the TTTS monitoring data is demonstrated using the monitored data IDX1 as an example.

5.1.1. Outliers removal

To construct a high-quality dataset for neural network training, all outliers in the data must be removed to accurately represent the normal behavior patterns of the structure. Fig. 16 displays the raw data of IDX1, where several outliers and missing data are clearly observed. According to the data preprocessing procedure shown in Fig. 11, we first detect extreme outliers using a boxplot. These extreme values are mostly caused by sensor malfunctions or data transmission issues, as indicated by red stars in Fig. 17. After eliminating the extreme outliers, the remaining outliers are further removed using a hybrid Sigma-LOF algorithm. To facilitate observation, data from May 8 and 9, 2022, were selected to demonstrate the outlier removal process. Firstly, the Sigma method is applied to detect outliers based on the 3σ criterion. The detection results are shown in Fig. 18(a), where red stars indicate detected outliers, and green diamonds represent undetected outliers. To further eliminate undetected outliers, strain data (IDX1) and relative temperature data (IDX2) were combined to form spatial data, and the LOF algorithm was subsequently applied for anomaly removal. The anomaly detection results are presented in Fig. 18(b), demonstrating that the LOF algorithm effectively identifies outliers that the Sigma method fails to detect. The spatial data is constructed by using relative temperature data as the X-axis and strain or tilt angle data as the Y-axis. If both the strain or tilt angle data and the corresponding relative temperature data exhibit anomalies, the cause is typically the data acquisition system rather than structural abnormalities, as temperature data generally does not undergo drastic changes within a short period. Conversely, if only the strain or tilt angle data shows sudden changes while the relative temperature data remains stable, it is more likely caused by structural abnormalities. Nevertheless, both types of anomalies are considered outliers. To build a high-quality dataset, all such outliers must be removed.

5.1.2. Missing value filling

Missing data imputation is performed using the NP algorithm. Fig. 19 presents a comparison between the filled data and the raw data. As shown in Fig. 19, the NP algorithm achieves satisfactory performance in filling the missing data.

5.1.3. Dataset construction

The monitoring data has been preprocessed according to the process described in Fig. 11. It is used to train the LSTM-MHSA-VAE neural network, in the form of a three-dimensional tensor of N × W × D, where N represents the batch size. W represents the time window size, and D represents the number of parameters, which is set to 17 in this study. For other types of structures, the dataset can be constructed by selecting relevant parameters such as strain or tilt angle from critical structural components, depending on the specific monitoring requirements. The dataset is divided into training and testing sets in a 4:1 ratio to ensure the robustness of model training and evaluation.

5.2. Evaluation of the LSTM-MHSA-VAE model

5.2.1. Hyperparameter optimization

In this study, a grid search approach was employed to systematically explore a wide range of hyperparameter combinations. Early stopping

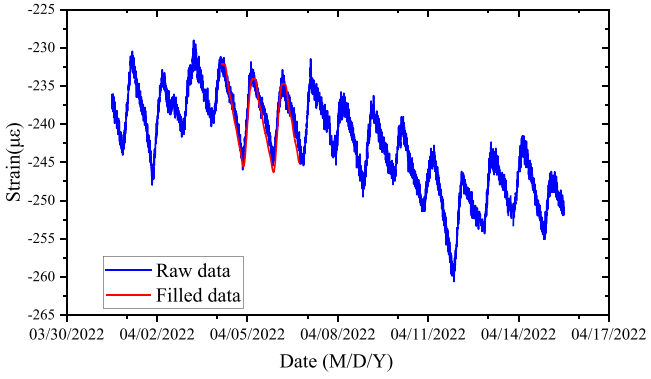
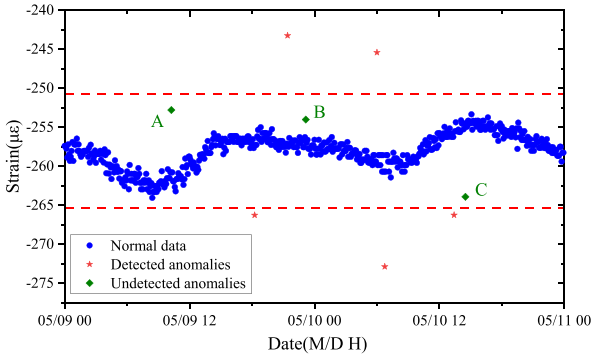
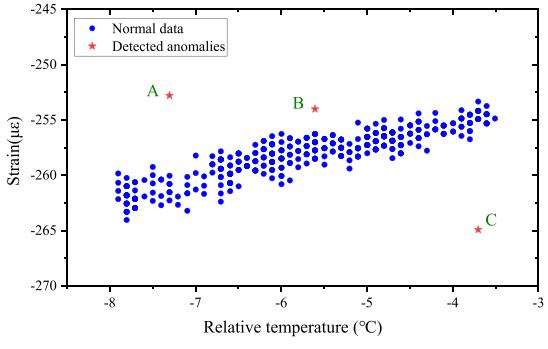


Fig. 19. Comparison of filled data and raw data.



(a) Sigma method



(b) LOF method

Fig. 18. Comparative results of TTTS data preprocessing.

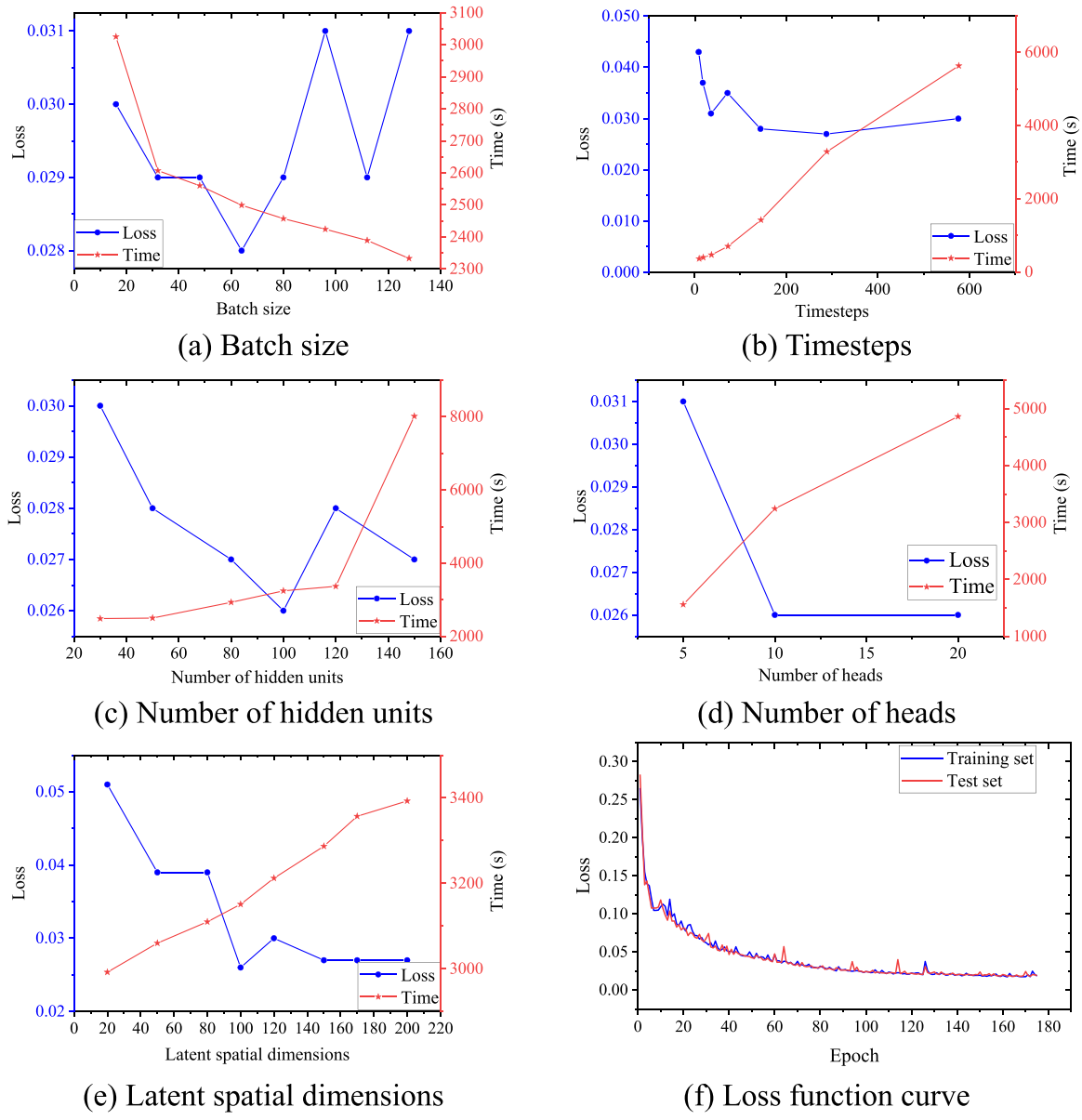


Fig. 20. Parameter optimisation and Loss function curve.

was applied to halt training if the test loss didn't improve for 10 consecutive epochs. To ensure robust evaluation of the model's performance, 5-fold cross-validation [42] was utilized to assess the model's generalization ability across different data splits.

The equipment used to carry out the hyperparameter optimization is CPU: 12th Gen Intel Core i7–12700H and GPU: NVIDIA GeForce RTX 3060 Laptop GPU. Hyperparameters include two categories: optimizer hyperparameters and model hyperparameters. Optimizer hyperparameters include learning rate, batch size, and optimizer type. The model hyperparameters depend on the type of model. The hyperparameters of the proposed LSTM-MHSA-VAE model can be classified into LSTM hyperparameters, MHSA hyperparameters, and VAE hyperparameters. The LSTM hyperparameters include the number of feature dimensions of the input data, the timesteps, and the number of hidden units. The MHSA hyperparameters include the number of attentional heads. The VAE hyperparameters include the number of layers, the dimensionality of the latent space, and the weight of the KL divergence term, denoted as β. To simplify the model structure, we employed a single-layer VAE model. The purpose of β is to balance the structure of the latent space and the quality of data reconstruction.

The optimizer hyperparameters are determined as follows:

- Learning rate: Empirically determined and set to 0.001.
- Batche size: The batch sizes evaluated in this paper are 16, 32, 48, 64, 80, 96, 112, and 128. After considering computational efficiency and model performance, a batch size of 64 was chosen, as shown in Fig. 20(a).
- Optimizer: Common optimizers for training deep learning models include Adam and RMSprop. Adam is particularly effective due to its adaptive learning rate, which helps manage gradient sparsity and non-smooth functions, enhancing training stability and convergence speed.

The hyperparameters for the model are determined as follows, with each hyperparameter described in sequence.

- Feature dimensions: The hyperparameter for feature dimensions is 17 because the constructed dataset includes 17 parameters.
- Timesteps: A larger timestep requires the model to have more memory to capture long-term dependencies, improving performance. However, as the number of steps increases, computational complexity also rises, potentially causing gradient vanishing or

Table 2
Comparative results of neural network performance.

NN	RMSE						MAE					
	1	2	3	4	5	mean	1	2	3	4	5	mean
AE	0.043	0.042	0.042	0.042	0.042	0.042	0.031	0.031	0.031	0.031	0.031	0.031
VAE	0.031	0.031	0.031	0.031	0.032	0.031	0.022	0.022	0.022	0.022	0.022	0.022
LSTM-AE	0.028	0.028	0.028	0.028	0.029	0.028	0.018	0.018	0.018	0.018	0.018	0.018
LSTM-VAE	0.025	0.026	0.024	0.025	0.026	0.025	0.015	0.015	0.014	0.015	0.016	0.015
LSTM- MHSA -VAE	0.018	0.019	0.019	0.020	0.019	0.019	0.010	0.011	0.011	0.011	0.011	0.011

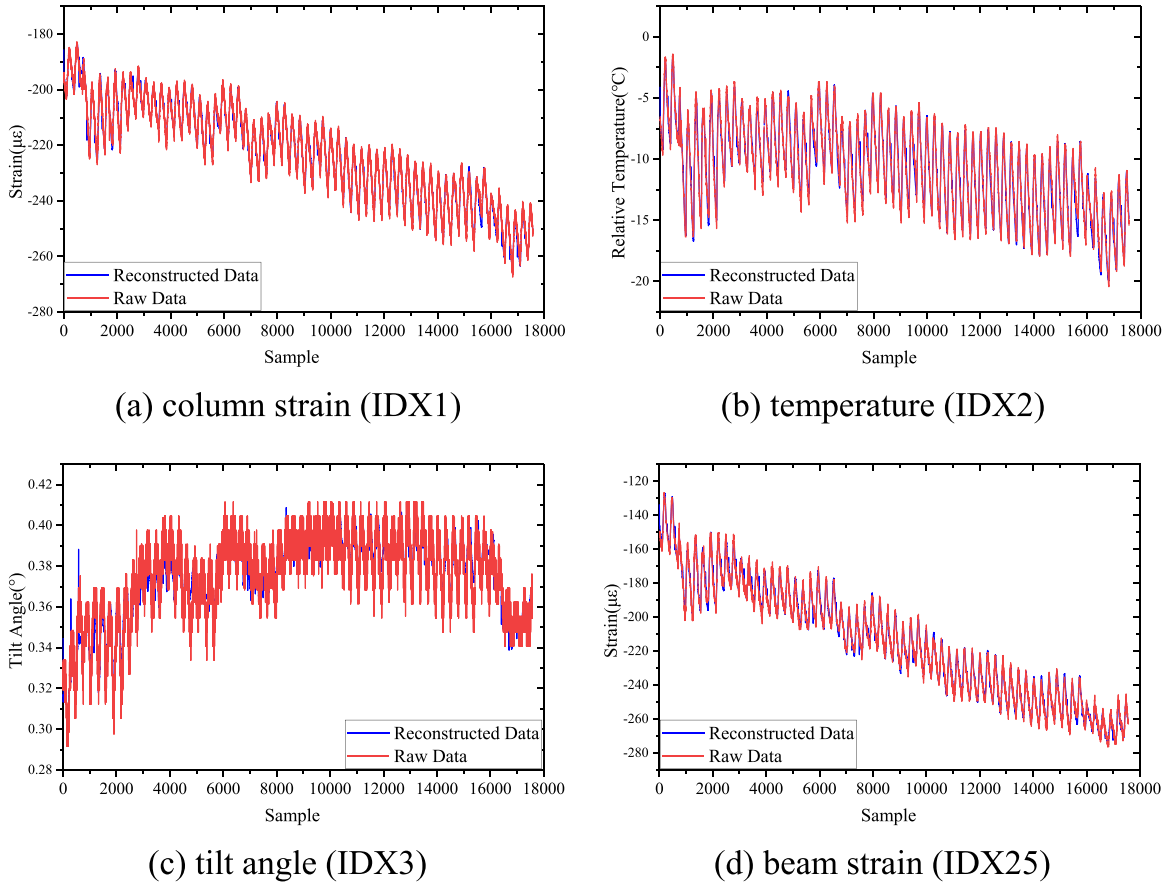


Fig. 21. Visualisation of LSTM- MHSA -VAE neural network reconstruction performance.

exploding issues. The optimal number of timesteps was found to be 288. The impact of timesteps on performance and training efficiency is shown in Fig. 20(b).

- Number of hidden units: Increasing the number of hidden units enhances the LSTM model's ability to learn and retain long-term dependencies [43]. However, too many hidden units can lead to wasted time and overfitting. The effect of hidden units on performance and time consumption is shown in Fig. 20(c).
- Number of heads: Increasing the number of heads improves the model's expressive ability but also increases training time and may cause overfitting. The possible options for the number of heads are 5, 10, and 20. After considering both performance and time, 10 heads were chosen. As shown in Fig. 20(d), setting the number of heads to 20 does not improve performance and wastes more time.
- Latent spatial dimensions: Optimal compression is crucial to avoid overfitting and excessive time consumption. A latent dimension of 100 is selected after careful analysis as showcased in Fig. 20(e).
- Weight β: The weight β balances the hidden space structure and data reconstruction quality. In this paper, the default value $\beta = 1$ is used.

The training and validation loss curves during the model training process are shown in Fig. 20(f). It can be observed that the model achieves steady convergence, with the training loss gradually decreasing and the validation loss stabilizing after a certain number of epochs. This indicates that the model effectively learns the underlying patterns of the data without significant overfitting.

5.2.2. *Neural network performance for data reconstruction*

Neural network performance is analyzed in two aspects: the performance of data reconstruction and the accuracy of outlier identification. The reconstruction and generalization ability of the model is assessed by comparing the MAE and RMSE between the raw data and the reconstructed data.

Table 2 presents the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) values for reconstructed TTTS data using five neural network architectures: AE, VAE, LSTM-AE, LSTM-VAE, and the proposed LSTM-MHSA-VAE. Notably, the LSTM-MHSA-VAE model achieves significantly lower RMSE and MAE values on the test dataset compared to other methods, demonstrating its superior reconstruction capability.

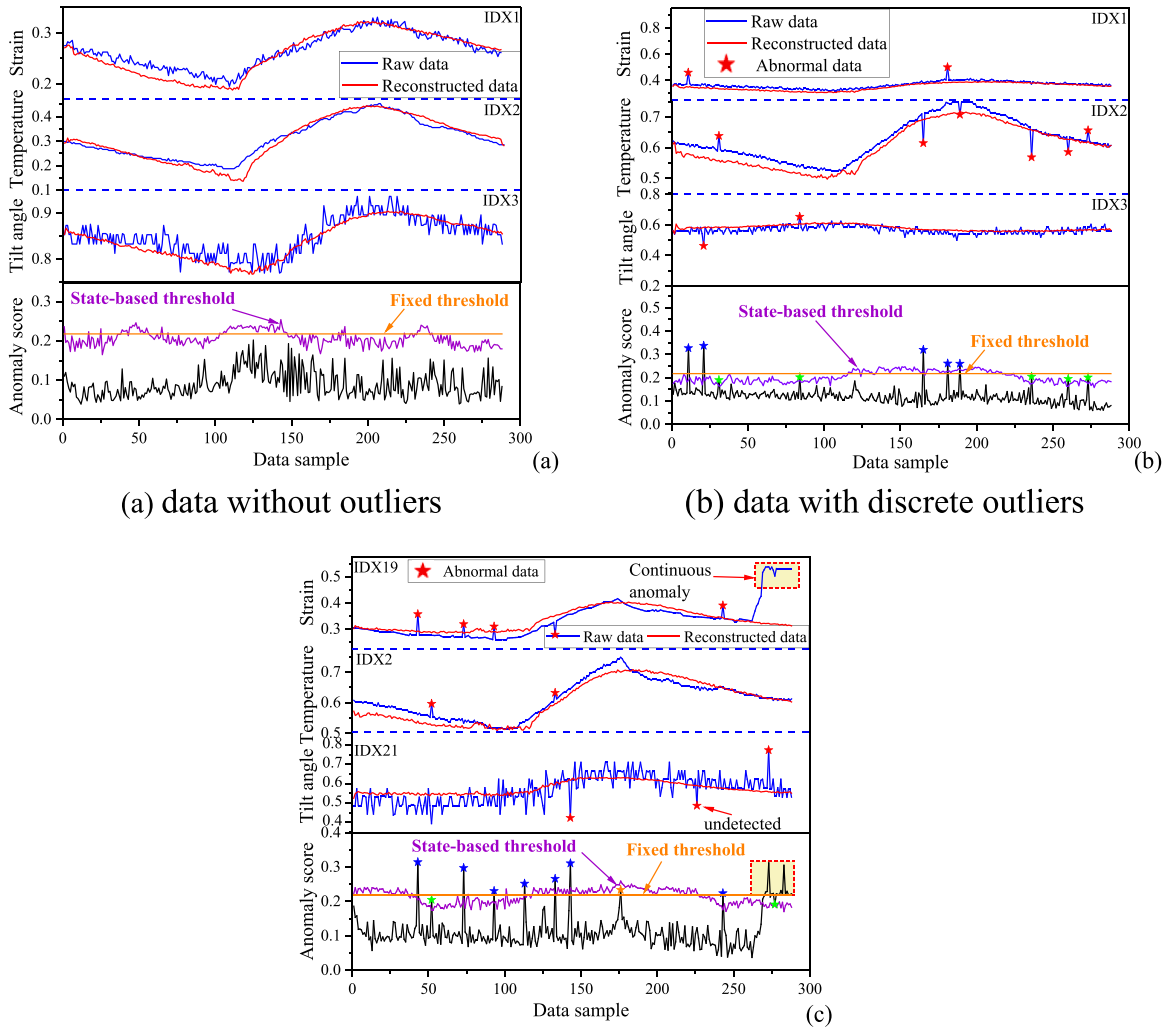


Fig. 22. Visualisation of the state-based threshold diagnostic approach.

To demonstrate the reconstruction performance of the LSTM-MHSA-VAE neural network, Fig. 21 provides a visual representation of the column strain data (IDX1), temperature data (IDX2), tilt angle data (IDX3), and beam strain data (IDX25). For each subplot, the raw data (in red) and the reconstructed data (in blue) are plotted against the sample index. The close alignment between the original and reconstructed data illustrates the model's ability to accurately capture the underlying patterns of the structural health monitoring signals. Furthermore, the discrepancy between the original and reconstructed data may indicate potential anomalies, which can be quantitatively analyzed by evaluating the reconstruction error.

5.2.3. *The accuracy of structural anomaly detection*

The performance of using state-based thresholds and fixed thresholds for detecting anomalous data using the TTTS monitoring data is studied. Section 4.3 describes the state-based thresholds for structural anomaly detection. The hidden state Z obtained from the LSTM-MHSA-VAE neural network is input into the trained anomaly score evaluator f_s to obtain state-based thresholds $\eta = f_s(Z) + c$. In this study, constant $c = 0$. Fixed thresholds are determined using statistical methods, specifically by the formula $\eta = \mu + k\sigma$, where μ is the sample mean and σ is the standard deviation. Assuming the data follows a normal distribution, approximately 99.7 % of the data points lie within three standard deviations from the mean. In this paper, we set k = 3, which is a commonly used value based on the 3-sigma rule in statistical anomaly detection to effectively capture significant deviations[36].

To illustrate the difference between state-based and fixed thresholds in detecting structural anomalies, monitoring data spanning one day (i.e., 288-time steps) is selected for analysis. Three types of monitoring data are analyzed in this section: data without outliers, data with discrete outliers, and data with continuous outliers. The anomalies are generated based on the following assumptions: 1) Discrete anomalies are randomly introduced into the dataset to simulate short-term, recoverable disturbances. Randomness ensures a realistic representation of unpredictable events. 2) Continuous anomalies are created by injecting a series of consecutive abnormal points over a predefined time interval, representing progressive structural damage or degradation. 3) The magnitude and frequency of anomalies are selected to cover a wide range of potential conditions that the monitoring system may encounter in real-world scenarios.

Fig. 22 shows the results of anomaly detection under the aforementioned scenarios. The upper three subplots in each figure display the raw and reconstructed data, while the bottom subplot shows the anomaly scores, the state-based thresholds, and the fixed thresholds calculated for each time step. Red stars in the figure indicate anomalies in the monitoring data, while blue stars represent anomaly scores exceeding both the state-based and fixed thresholds. Green stars indicate anomaly scores above the state-based thresholds but below the fixed

Table 3
Summary of results.

Dataset	Model Type	State threshold			Fixed threshold		
		precision	recall	F1 score	precision	recall	F1 score
1	LSTM-AE	0.83	0.79	0.81	0.76	0.81	0.78
2		0.82	0.80	0.81	0.77	0.79	0.78
3		0.85	0.79	0.82	0.76	0.78	0.77
mean		0.83	0.79	**0.81**	0.76	0.79	**0.78**
1	LSTM-VAE	0.87	0.82	0.84	0.80	0.78	0.79
2		0.88	0.81	0.84	0.81	0.77	0.79
3		0.85	0.80	0.82	0.79	0.78	0.78
mean		0.87	0.81	**0.84**	0.80	0.78	**0.79**
1	LSTM-MHSA-VAE	0.93	0.91	0.92	0.80	0.78	0.79
2		0.91	0.92	0.91	0.85	0.84	0.84
3		0.92	0.90	0.91	0.87	0.88	0.87
mean		0.92	0.91	**0.91**	0.84	0.83	**0.84**

thresholds, and orange stars represent anomaly scores above the fixed thresholds but below the state-based thresholds. The first three types of stars correspond to correctly identified anomalies, while the last type refers to incorrectly identified anomalies. In Fig. 22(a), no anomalies are detected in the good data. However, it is noteworthy that the state-based threshold fluctuates over time, which is correlated with the quality of the reconstructed data, i.e., the hidden state **Z**. Fig. 22(b) shows that state-based thresholds are more capable of detecting outliers that exhibit subtle deviations from the normal data trend, compared to fixed thresholds. Fig. 22(c) illustrates that the state-based threshold is particularly effective in identifying continuous outliers, while the orange star highlights an incorrect anomaly detection using fixed thresholds. In summary, state-based thresholds prove to be more effective in detecting outliers than fixed thresholds.

To verify the robustness and accuracy of the proposed LSTM-MHSA-VAE neural network with state-based threshold for outlier detection, criteria like precision, recall, and F1 score as discussed in Section 3.3.2 are adopted. The time span of the three datasets covers 31 days (i.e., 8928-time instants) in the months of May, July, and August 2023. Considering that there are few outliers in the raw monitoring data, 200 outliers are randomly generated in each dataset. The dataset is then fed into the trained neural network and anomaly identification is performed on the outlier data using state-based thresholding and fixed thresholding. Table 3 summarizes the anomaly detection results of LSTM-AE, LSTM-VAE, and LSTM-MHSA-VAE neural networks using both fixed thresholds and state-based thresholds. It can be observed that: 1) The state-based threshold LSTM-MHSA-VAE neural network achieves higher accuracy, in general, in anomaly detection compared to other neural networks; 2) The state-based threshold method significantly improves the anomaly detection accuracy compared to the fixed threshold method; and 3) The state-based threshold LSTM-MHSA-VAE neural network has the F1 scores above 0.90, in general, which is better than those from the other two methods indicating good robustness in anomaly detection.

5.2.4. *Anomaly detection cases*

The anomaly detection method proposed in this paper has been successfully integrated into the SHM system, achieving real-time monitoring and early warning for the TTTS. From February 1–7, 2024, maintenance and reinforcement work was conducted on the monitored structure. The construction activities impacted the structural components, causing the tilt angle and strain data of timber columns and beams to deviate from their original patterns.

Fig. 23 presents the monitoring results over these 7 days, including four tilt angle data series, six strain data series, and a comparison of anomaly scores with state-based thresholds. From Fig. 23(a), it can be observed that tilt angle data fluctuated significantly during the daytime due to construction disturbances. Notably, IDX9 experienced two sudden changes on February 4 and 5, while IDX21 underwent a sudden change on February 4 and did not return to its original state, instead stabilizing at a new equilibrium. In Fig. 23(b), the strain data for IDX7 and IDX19 exhibited multiple abrupt changes, which did not revert to the normal state but rather stabilized at new levels. These findings indicate that the structural components' responses to disturbances can be classified into the following two scenarios: Rapid Recovery: Data experiences abrupt changes and quickly returns to normal state. State Shift: Data deviates from the original pattern and maintains stability at a new equilibrium.

Fig. 23(c) illustrates the identification of these anomalies. For

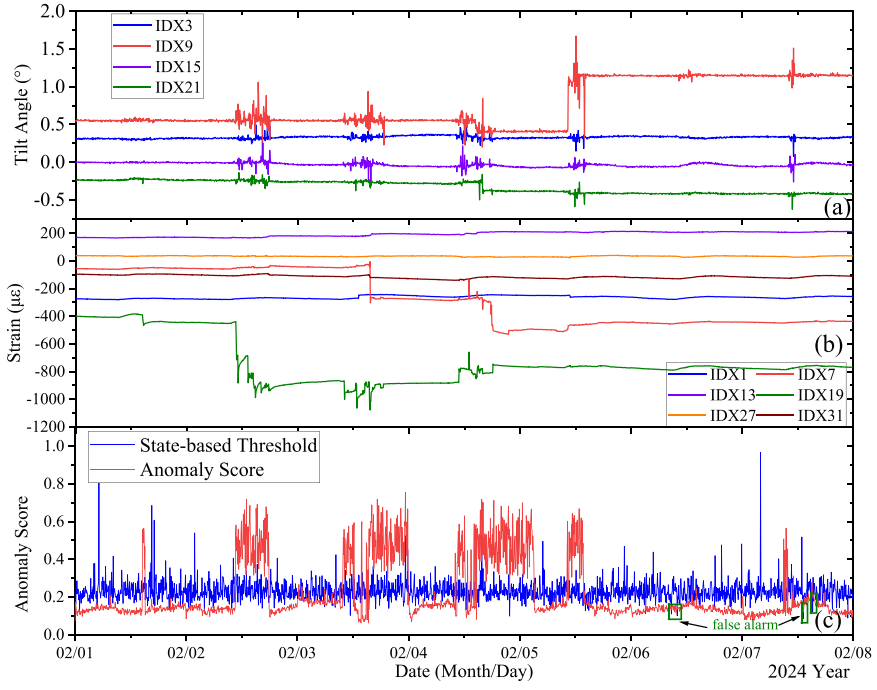


Fig. 23. Demonstration of anomaly detection performance in practical engineering applications.

Scenario 1, anomaly scores exceed the threshold momentarily and quickly return to normal. For Scenario 2, anomaly scores remain above the threshold for a prolonged period. It is worth noting that prolonged anomaly alarms suggest that the structure may have undergone a state shift, which should be closely monitored and carefully analyzed.

The proposed method demonstrates good performance in practical applications, accurately identifying anomalies caused by construction activities. However, due to the current model's reconstruction capability, false alarms may occur in certain situations. Future research will focus on improving this method by exploring more suitable models and anomaly diagnosis techniques to enhance detection accuracy and robustness. In particular, adaptive mechanisms will be developed to better recognize new equilibrium states during structural state shifts, effectively reducing false alarms and providing more reliable support for practical engineering applications.

6. Conclusions

The following conclusions may be drawn:

(1) Feature analysis was conducted on the TTTS monitoring data, which exhibits both long-term and short-term dependencies. The data also shows clear daily and annual periodicities, with its distribution changing over time.

(2) A data preprocessing framework is proposed which includes the sigma-LOF algorithm for anomaly detection and the NP algorithm for missing data imputation.

(3) A LSTM-MHSA-VAE neural network is proposed for data reconstruction. Compared to traditional AE, VAE, LSTM-AE, and LSTM-VAE neural networks, improved performance in data reconstruction is achieved.

(4) This paper proposes a state-based threshold determination method. Compared to existing methods, the state-based threshold LSTM-MHSA-VAE neural network demonstrates higher accuracy and better robustness in structural anomaly detection.

(5) To provide more physical meaning to the proposed strategy, further exploration of the physical relationships between the monitoring data features of TTTS and the structural properties represented by the TTTS data is required.

CRediT authorship contribution statement

Yang Na: Writing – review & editing, Resources, Project administration, Funding acquisition. **Qian Xiaowen:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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